Yiyang Wang

Ph.D. Candidate in Computer Science

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Research Profile

Research in computer systems and performance, focused on caching, memory locality, stochastic workload modeling, and AI-system memory behavior. Current work spans analytical cache modeling, online caching algorithms, GPU embedding-memory optimization, and locality-based analysis of LLM workloads.

Education

University of Rochester — **Aug 2023–May 2028 (anticipated)**
Ph.D. in Computer Science; advisor: Chen Ding — *Rochester, NY*

University of Rochester — **Aug 2022–May 2023**
M.S. in Computer Science — *Rochester, NY*

New York University — **Jan–Jun 2022**
Tandon Bridge Program for Computer Science, Certificate of Distinction — *New York, NY*

Peabody Institute of Johns Hopkins University — **Aug 2019–May 2021**
M.M. in Composition — *Baltimore, MD*

Reed College — **Aug 2014–May 2018**
B.A. in Music, Phi Beta Kappa — *Portland, OR*

Publications and Manuscripts

Ranking Human and LLM Texts Using Locality Statistics 2026
Yiyang Wang, Chen Ding, and Hangfeng He. Findings of EACL 2026, pp. 5337–5348.
- Applied reuse distance and Data Movement Distance to human- and LLM-generated text with statistical and machine-learning analysis, producing locality features that outperform frequency-only methods.

Continuous-Time Modeling of Zipfian Workload Locality 2025
Yiyang Wang, Chen Ding, Leo Sciortino, and Linlin Chen. ACM TOMPECS, vol. 10, no. 4.
- Developed continuous-time numerical models and an $O(\log N)$-group partitioning method for LRU on 10-million-object Zipfian workloads, reducing multi-day simulation to millisecond-scale prediction at approximately 0.1% miss-ratio error.

Modeling and Practical Realization of Optimal Online Caching for Stochastic Workloads
Yiyang Wang, Chen Ding, and Leo Sciortino. Under review; preprint available upon request.
- Proved Uniform Eviction-Free Lease (EFL) converges to static LFU as leases grow under a steady-state flow-balance approximation; developed a finite-lease solver $10^4\times$ faster than simulation with absolute miss-ratio error below 0.005 in Uniform EFL validation. Designed Nursery EFL, reducing the mean miss-ratio gap to LFU by 60–74% versus the strongest evaluated online baseline at each skewness on stationary synthetic Zipfian workloads.

Structure-Aware Representation Learning for Symbolic Music with Graph Neural Networks
Huiran Yu, Wei-Cheng Lin, Yiyang Wang, and Zhiyao Duan. Under review.
- Built MuseScore/QML annotation tools and coordinated development of a cross-genre symbolic-music dataset, enabling evaluation of structure-aware graph-neural-network representations.

Research Experience

University of Rochester — **Aug 2023–Present**
Research Assistant, Systems Research Group — *Rochester, NY*
- Lead research across cache management, workload modeling, memory locality, and AI-system performance, integrating analytical, simulation-based, and empirical methods.
- Built and maintain reusable high-performance Rust simulation infrastructure across multiple caching projects, standardizing workload generation, policy evaluation, and compact expiration and cleanup structures.

University of Rochester — **Jan 2023–May 2024**
Research Assistant, Audio Information Research Lab — *Rochester, NY*
- Developed self-supervised VAE and contrastive-learning pipelines in PyTorch and Fairseq for musical-pattern retrieval and generation experiments.

Ongoing Research Projects

GPU Embedding Retrieval and Memory Optimization — *CUDA, NVIDIA Nsight Compute* **Ongoing**

- Extended CUDA embedding benchmarks and automated A100/H100 sweeps across batch size, embedding width, Zipfian skew, and table layout, characterizing layout-dependent L1/L2/DRAM behavior.
- Investigate hot-row placement, L2 persistence, and CPU–GPU caching and offloading strategies to improve embedding-memory behavior.

Locality-Aware Context Management for LLMs — *locality, vLLM, Lean proof generation* **Ongoing**

- Develop recency- and reuse-aware policies that move information among verbatim, compressed, summarized, and archived states under fixed token and KV-cache budgets.
- Built a long-context Lean proof-generation evaluation pipeline comparing locality-aware policies with full-context and fixed-window baselines across proof success, context size, compression ratio, and token cost.
- Built and configured reproducible vLLM inference infrastructure with batching, GPU/server configuration, request-level seeding, and speculative decoding/MTP; evaluation is ongoing across task quality, throughput, and inference overhead.

Professional Experience

NYC Music Services **Dec 2021–Feb 2022**

Contract Software Developer *New York, NY*

- Designed and implemented a cross-platform application in Python, wxPython, and pdfrw, automating PDF layout, metadata, and batch file operations for professional sheet-music production.

Peabody Institute of Johns Hopkins University **2021–2022**

Graduate Assistant, Computer Music Department *Baltimore, MD*

- Maintained computer-music studios and hardware/software systems and coached students, supporting reliable lab access and course delivery.

Selected Technical Projects

Concurrent Lease Cache — *Rust, concurrent systems* **2025**

- Implemented a concurrent lease-cache simulator in Rust with fine-grained hash locks and lock-efficient synchronization, improving throughput by up to 2.4× over the serial implementation.

Idiom Paraphrasing via Contrastive Learning — *BART, PyTorch* **2023**

- Fine-tuned BART in PyTorch with random masking and idiom–literal contrastive supervision, improving BLEU, ROUGE, and METEOR scores while generalizing to unseen idioms.

Jazz Machine Co-Improviser — *RNN, real-time audio* **2022–2023**

- Developed an RNN-based real-time audio agent that generated jazz-trio accompaniment and chordal cues from live performance input, culminating in a 2023 symposium poster and demonstration.

Distributed Version Control System — *Rust* **2022**

- Led a four-person team building a Git-like command-line system in Rust and implemented its filesystem-facing backend, enabling local versioning through a modular command interface.

Selected Creative Work

Mind Your Step — *Interactive music installation* **2022**

- Created a movement-responsive mobile sound application and multichannel acousmatic environment, turning participant motion, footsteps, and device audio into an interactive installation.

Unread — *Interactive multimedia composition* **2020–2021**

- Built an interactive system that converted audience text messages into musical Morse code, integrating live participation with a piano-and-percussion ensemble.

Posters and Presentations

Time-Continuous Modeling of Zipfian Workload Locality **2024**

Poster and presentation, ACM SIGMETRICS Student Research Competition

An Interactive Computational System to Accompany Jazz Improvisation **2023**

Poster and demo, Graduate Research Symposium, University of Rochester

Teaching and Service

University of Rochester Teaching Center **Summer 2025; Winter and Summer 2026**
Graduate Teaching Assistant Coach *Rochester, NY*

- Trained new graduate TAs across STEM departments in instructional policy, grading, communication, and classroom management, improving readiness for teaching and lab assignments.
- Designed and led interactive pedagogy sessions adopted in the university's recurring graduate-TA orientation program, expanding reusable training capacity.

University of Rochester **Aug 2023–May 2025**
Teaching Assistant, Department of Computer Science *Rochester, NY*

- Led workshops and review sessions for programming languages, computability, and collaborative software design, reinforcing difficult course concepts.
- Coordinated grading, feedback, office hours, and undergraduate TA teams, improving consistency across course sections.

Reed College **2015–2018**
Music Theory Tutor *Portland, OR*

- Delivered one-on-one music-theory coaching, adapting explanations and exercises to individual learning needs.

Private Piano Studio **2016–2018**
Piano Teacher *Portland, OR*

- Taught children and adults with varied prior experience, tailoring repertoire and technique instruction to individual goals.

Honors and Awards

Icarus Quartet iQ Tests Commissioning Contest 2020
Artistic Excellence Award, Peabody Institute 2019–2021
Phi Beta Kappa, Reed College 2018
Locher Creative Scholarship, Reed College 2017
Wintergreen Composer's Scholarship, Wintergreen Summer Music Festival 2017
Grand Prize, Chamber Music Northwest Composition Competition 2016
Rothchild Summer Stipend for Music Study, Reed College 2016

Skills

Programming: Rust, Python, C++, C, CUDA, Java, Bash, JavaScript, SQL, OCaml, Rocq/Coq, Lean 4
Machine Learning and NLP: PyTorch, Lightning, Hugging Face Transformers, Fairseq, vLLM, spaCy, scikit-learn, Weights & Biases
Systems and Performance: Linux, Git, NVIDIA Nsight Compute, perf, Slurm
Data and Visualization: NumPy, pandas, Matplotlib
Music Technology: Max/MSP, SuperCollider, OpenMusic, Csound, Processing, Tone.js, music21, MuseScore/QML
Audio Production: Logic Pro, Reaper, Ableton Live; recording, mixing, studio interfaces, multitrack microphone setups